

Letter to the Editor

Re: Structured Educational Interventions and Postoperative Outcomes

To the Editor,

We read with great interest the study evaluating the impact of structured educational interventions on postoperative outcomes. The topic is clinically important, and the authors should be appreciated for addressing perioperative patient education and recovery outcomes. However, several methodological limitations should be considered while interpreting the findings.

1. Generalizability and Statistical Power

The study was conducted at a single center with a relatively small sample size, which may limit the generalizability of the results to broader patient populations and different healthcare settings. In addition, the limited sample size may have reduced the statistical power of the study and increased the risk of type II error.^[1]

2. Selection Bias and Confounding

Randomization could not be implemented because of institutional room-assignment procedures. This may have introduced selection bias and baseline imbalance between the study groups. Furthermore, differences in patients' educational backgrounds may have influenced their understanding of perioperative instructions and postoperative recovery behaviors, thereby acting as potential confounding factors affecting the observed intervention outcomes.^[2]

3. Intervention Fidelity and Comprehension Assessment

Although the intervention group received a structured educational session with brochure-based counseling, no formal assessment of patient comprehension or intervention fidelity was performed. The absence of teach-back methods or knowledge-retention assessments makes it difficult to confirm whether patients adequately understood and retained the provided information. Therefore, some of the observed benefits may have resulted from increased healthcare-provider interaction and attention rather than the educational intervention itself.^[3]

4. Absence of Blinding and Outcome Bias

The apparent lack of blinding among healthcare personnel and outcome assessors may have introduced both performance and detection bias. Patients in the intervention group received greater interaction with the clinical team, which itself may have influenced subjective outcomes such as pain perception, comfort, and recovery experience. Moreover, several study endpoints, including Visual Analog Scale scores and comfort-related measures, were observer- and patient-dependent outcomes that are vulnerable to expectancy effects in the absence of

blinding. Previous evidence suggests that lack of blinding may exaggerate treatment effects, particularly in studies involving subjective clinical outcomes.^[4]

Despite these limitations, the study contributes valuable insight into the role of perioperative educational interventions in improving postoperative recovery. Further multicenter randomized controlled trials with larger sample sizes, standardized educational assessments, and appropriate blinding procedures are recommended to strengthen the evidence in this area.

References

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